`bispectrum`: Selective G-Bispectra Made Practical

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Abstract

Write abstract ($\sim$150 words). Key points to hit:

- We release `bispectrum`, the first PyTorch library for computing selective G-bispectra—the unique complete and exact group-invariant features, with $O(|G|)$ coefficients for finite groups.
- The library supports 7 group/domain pairs (C_n, D_n, O, $\mathrm{SO}(2)/S^1$, $\mathrm{SO}(2)/\text{disk}$, $\mathrm{SO}(3)/S^2$, torus) as differentiable `nn.Modules` with GPU acceleration, signal inversion, and applications to invariant pooling, orbit-invariant regularization, and multi-reference alignment.
- Building the library required solving two open engineering challenges: bridging steerable CNN features to group Fourier coefficients for non-abelian groups (Theorem 1), and resolving a previously undocumented inversion failure for the octahedral group via bootstrap + Levenberg–Marquardt.
- On PatchCamelyon, bispectral pooling achieves the highest AUC among equivariant nonlinearities, the most stable accuracy–parameter Pareto frontier across architectures, and near-perfect rotation invariance. On RotMNIST, a bispectral contrastive regularizer improves rotation robustness.
- The library is pip-installable, MIT-licensed, and integrates into standard PyTorch training pipelines.

1 Introduction

Write introduction ($\sim$1 page). Follow the narrative arc below.

The invariant nonlinearity bottleneck. Equivariant convolutional neural networks [Cohen and Welling, 2016, Weiler and Cesa, 2019] achieve symmetry by design, replacing data augmentation with group convolutions that guarantee equivariant feature maps. Yet every such architecture must eventually produce *invariant* features—for classification, regression, or any prediction that should not depend on input orientation. The choice of invariant map is a critical design decision with no consensus solution (Table 1). Existing approaches sacrifice either completeness (norm, gate), exactness (Fourier pointwise), or efficiency (tensor product, full bispectrum).

The gap from theory to practice. The selective G-bispectrum [Mataigne et al., 2024] provides $O(|G|)$-cost complete invariants for finite groups, and Myers and Miolane [2025] extend this to the disk. However, no practical software exists. Moving from theorems to a usable library required solving three problems:

1. No GPU-accelerated implementation exists for any group beyond toy examples.

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Table 1: Invariant nonlinearities for equivariant CNNs. The selective bispectrum, implemented in our library, is the only method that is simultaneously exact, complete, and linear-cost for finite groups.

Nonlinearity	Exact?	Complete?	Coefficients	Key limitation
Norm	Yes	No	$O(\|G\|)$	Discards phase
Gate	Yes	No	$O(2\|G\|)$	Scalar bottleneck
Fourier pointwise	No	Yes	$O(\|G\| \log \|G\|)$	Aliasing
Tensor product	Yes	Yes	$O(\|G\|^3)$	Cubic cost
Full bispectrum	Yes	Yes	$O(\|G\|^2)$	Quadratic cost
Sel. bispectrum	**Yes**	**Yes**	$\mathbf{O}(\|\mathbf{G}\|)^\dagger$	**(this work)**

†For finite groups where selectivity is proven [Mataigne et al., 2024]. Selective $\mathrm{SO}(3)$ on S^2 remains open.

2. For the octahedral group, the theoretically prescribed bootstrap inversion algorithm fails due to an $\mathrm{SO}(3)$ phase ambiguity not addressed by the Clebsch–Gordan structure (Section 3.5).
3. Steerable CNN feature maps live in intertwiner spaces, not in group Fourier spaces—for non-abelian groups, one cannot naïvely plug steerable outputs into the bispectrum formula (Section 3.4).

Contributions. We present `bispectrum`, a PyTorch library that makes selective G-bispectra practical as differentiable primitives for invariant feature extraction, regularization, and signal reconstruction:

1. **`bispectrum` library** (Section 3): The first PyTorch library for selective G-bispectra, supporting 7 group/domain pairs as `nn.Modules` with GPU acceleration, signal inversion, and seamless integration into training pipelines. Beyond invariant pooling, the library enables orbit-invariant losses, metric learning, and multi-reference alignment (Section 3.3). Pip-installable, MIT-licensed, with CI/CD and >90% test coverage.
2. **Engineering contribution: steerable-to-Fourier bridge** (Section 3.4): We prove conditions under which steerable convolution outputs yield valid group Fourier coefficients for bispectral computation (Theorem 1), enabling integration with existing equivariant architectures.
3. **Engineering contribution: octahedral inversion** (Section 3.5): We identify a previously undocumented inversion failure for the octahedral group and resolve it via bootstrap initialization followed by Levenberg–Marquardt refinement.
4. **Empirical validation** (Section 4): On PatchCamelyon, bispectral pooling achieves the highest AUC among equivariant nonlinearities, the flattest accuracy–parameter Pareto frontier (most robust to architecture choices), and near-perfect rotation invariance ($\sigma_{\text{AUC}} < 0.002$). On RotMNIST, a bispectral contrastive regularizer improves rotation robustness without augmentation labels.

   Update RotMNIST claim once experiment is run.

2 Mathematical Preliminaries

2.1 Equivariant CNNs and invariant maps

Write ∼0.5 page on equivariant CNNs. Cover:

- Group convolutions: a convolution layer is G-equivariant if $[\pi(g) \circ \Phi](f) = \Phi(\pi(g)f)$ for all $g \in G$. Cite Cohen and Welling [2016].
- Regular vs. steerable representations. In regular representations, features are functions $f : G \to \mathbb{R}^C$. In steerable representations, features transform as $f \mapsto \rho(g)f$ for a representation ρ [Weiler and Cesa, 2019].

- The need for G-invariant pooling: to produce predictions that do not depend on the input orientation, one must map equivariant features to invariant ones.
- Existing invariant maps: norm (discards phase), gated nonlinearity (scalar bottleneck), Fourier pointwise (approximate equivariance due to aliasing [Franzen and Wand, 2021]), tensor product / CG product ($O(L^3)$ cost [Kondor et al., 2018]).

2.2 The G-bispectrum

Write ∼0.5 page. Merge the G-bispectrum definition and selective bispectra into one subsection. Cover:

- Group Fourier transform: for a finite group G with irreducible representations $\{\rho_i\}_{i\in\hat{G}}$,

$$\hat{F}(\rho_i) = \sum_{g\in G} f(g)\,\rho_i(g) \in \mathbb{C}^{d_i\times d_i}. \tag{1}$$

- The G-bispectrum: given the Clebsch–Gordan decomposition $\rho_i \otimes \rho_j \cong \bigoplus_k m_{ij}^k \rho_k$ with intertwining matrix C_{ij},

$$\beta_{\rho_i,\rho_j} = C_{ij}^\top\,(\hat{F}(\rho_i)\otimes\hat{F}(\rho_j))\,C_{ij}\,\hat{F}(\rho_k)^\dagger. \tag{2}$$

- Completeness: the bispectrum is a *complete* invariant—it separates orbits under the G-action [Kakarala, 1992].
- Full cost: $O(|G|^2)$ coefficients for finite groups.
- **Selective bispectra** [Mataigne et al., 2024]: for finite groups, $O(|G|)$ bispectral coefficients suffice for completeness, selected via a BFS on the Kronecker product table of $\hat{G}$. The disk extension [Myers and Miolane, 2025] gives $O(N)$ Fourier–Bessel coefficients for $\mathrm{SO}(2)$-invariance on the unit disk.
- Bootstrap inversion algorithms recover f (up to group action) from the selective bispectrum for commutative and dihedral groups.

2.3 The representation-theoretic gap

Write ∼0.3 page briefly identifying the steerable gap. This motivates Section 3.4. Cover:

- Steerable convolution outputs transform as $f \mapsto \rho(g)f$ where ρ is a (potentially reducible) representation.
- For abelian groups (C_n, $\mathrm{SO}(2)$): all irreps are 1D characters, so intertwiner coefficients *are* Fourier coefficients. No gap.
- For non-abelian groups (D_n, O): the intertwiner space $\mathrm{Hom}_G(\rho_i, \rho_{\text{out}})$ has dimension m_i = multiplicity of ρ_i in ρ_{out}. When $m_i < d_i$ (the dimension of ρ_i), steerable outputs do not span the full Fourier matrix—one cannot naïvely apply the bispectrum formula.
- Concrete example: for D_4 with output type $\rho_1 \oplus \rho_0$, the steerable features give only 3 numbers but the Fourier transform of ρ_1 needs $2^2 = 4$.
- We solve this within the library (Section 3.4).

3 The `bispectrum` Library

This section describes the design, scope, and key engineering contributions of the `bispectrum` library. Following the structure advocated by Paszke et al. [2019] and Gardner et al. [2018], we focus on *design principles and trade-offs* rather than a usage tutorial.

3.1 Design principles

Write ∼0.5 page on design principles. Frame as a *design manifesto* (per PyTorch Reviewer 2's advice). Cover:

- **P1: Naming encodes the math.** Class names follow `{Group}on{Domain}` (e.g., `CnonCn`, `SO3onS2`, `SO2onDisk`). A reader unfamiliar with the codebase immediately identifies the group and domain from the class name. This is deliberately mathematical rather than verbal (`CyclicBispectrum`) because the mathematical name carries precise meaning and scales as the library grows.
- **P2: `nn.Module` only.** Every bispectrum computation is a `torch.nn.Module` subclass. No parallel functional API. One way to compute a bispectrum. Precomputed quantities (CG matrices, DFT kernels, Bessel roots) are registered as non-learnable buffers, so `.to(device)` and `.to(dtype)` work transparently.
- **P3: Selective by default.** All modules default to `selective=True`, computing the minimal $O(|G|)$ coefficient set. The full $O(|G|^2)$ bispectrum is available via `selective=False` for debugging and comparison.
- **P4: Raw signals in.** Modules accept real-valued signals in the natural domain (spatial grid, discrete cycle, sphere) and handle the Fourier transform internally. For example, `SO3onS2` accepts signals of shape $(B, n_{\text{lat}}, n_{\text{lon}})$ and computes the spherical harmonic transform internally. This keeps the interface clean and prevents Fourier-convention mismatches.
- **P5: Inversion on-module.** Every module exposes `bsp.invert(beta)` that returns the reconstructed signal (up to group-action indeterminacy) or raises a precise `NotImplementedError` with guidance.
- **P6: Minimal dependencies.** The library depends only on `torch` and `numpy`. No external CG library: Clebsch–Gordan matrices for D_n are computed analytically, for $\mathrm{SO}(3)$ they are bundled as pre-computed JSON, and Bessel roots for the disk are computed via a pure-torch Newton-bisection solver using the interlacing property.

3.2 Supported groups and API

Table 2 lists all 7 group/domain pairs currently supported by the library.

Table 2: Supported group/domain pairs in the `bispectrum` library. Selective coefficient counts are $O(|G|)$ vs. $O(|G|^2)$ for the full bispectrum. All modules share a uniform `nn.Module` interface.

Module	Group / Domain	Selective	Full	Inversion	Complexity
`CnonCn`	C_n on C_n	n	n^2	✓	$O(n \log n)$
`SO2onS1`	$\mathrm{SO}(2)$ on S^1	n	n^2	✓	$O(n \log n)$
`TorusOnTorus`	$\prod C_{n_l}$ on torus	$\lvert G\rvert$	$\lvert G\rvert^2$	✓	$O(\lvert G\rvert \log \lvert G\rvert)$
`DnonDn`	D_n on D_n	$\sim 4\lvert D_n\rvert$	—	✓	$O(n)$
`OctaonOcta`	O on $\mathbb{R}^3$	172	—	✓*	$O(1)$
`SO3onS2`	$\mathrm{SO}(3)$ on S^2	—	$O(L^2)$	—	$O(L^2 d_{\max}^2)$
`SO2onDisk`	$\mathrm{SO}(2)$ on disk	N	N^3	✓	$O(N)$

*Bootstrap + LM refinement (Section 3.5).

Module interface contract. Every module satisfies: (i) `forward(f)` accepts a real-valued signal of shape $(B, \ldots)$ and returns a complex tensor of shape $(B, \texttt{output_size})$; (ii) `output_size` gives the number of bispectral coefficients; (iii) `index_map` maps each flat output index to its irrep index tuple (ρ_i, ρ_j); (iv) no trainable parameters; (v) deterministic output.

Usage examples.

```python
import torch
from bispectrum import CnonCn, SO3onS2, SO2onDisk

# Cyclic group on Z/8Z: selective reduces 64 -> 8 coefficients
bsp = CnonCn(n=8, selective=True)
f = torch.randn(32, 8) # batch of 32 signals
beta = bsp(f) # (32, 8), complex64
f_rec = bsp.invert(beta) # (32, 8), up to cyclic shift

# SO(3) on the sphere
bsp_s2 = SO3onS2(lmax=5, nlat=64, nlon=128)
```

```
f_sphere = torch.randn(4, 64, 128) # spherical signal
beta_s2 = bsp_s2(f_sphere) # (4, output_size), complex64

# SO(2) on the unit disk
bsp_disk = SO2onDisk(L=16, selective=True)
f_disk = torch.randn(4, 16, 16) # disk image
beta_disk = bsp_disk(f_disk) # (4, N), complex128
f_rec = bsp_disk.invert(beta_disk) # (4, 16, 16)
```

3.3 Beyond invariant pooling

Because every module is a differentiable `nn.Module` with both `forward` and `invert`, the library supports use cases beyond invariant feature extraction:

Orbit-invariant regularization. For generative or reconstruction tasks, the bispectrum provides a rotation-invariant loss without augmentation: the loss $\|\beta^{\text{sel}}(f_{\text{pred}}) - \beta^{\text{sel}}(f_{\text{target}})\|^2$ is invariant to the G-action on both signals by construction. We demonstrate this as a contrastive regularizer in Section 4.5.

Orbit distance and metric learning. $\|\beta^{\text{sel}}(f) - \beta^{\text{sel}}(g)\|$ defines a distance between signals modulo group action, suitable for retrieval, clustering, or nearest-neighbor classification on equivalence classes.

Multi-reference alignment. Recovering a signal from noisy copies at unknown orientations reduces to bispectral averaging followed by inversion [Myers and Miolane, 2025, Bendory et al., 2017]. The `invert()` method provides this primitive.

Symmetry detection. Vanishing patterns in bispectral coefficients reveal hidden symmetries in data: a signal invariant to a subgroup $H \leq G$ produces a bispectrum with predictable zero blocks.

```
import torch
from bispectrum import CnonCn

bsp = CnonCn(n=8, selective=True)

# 1. Invariant features for classification
features = bsp(signal) # (B, 8), complex64

# 2. Orbit-invariant regularization loss
loss = torch.nn.functional.mse_loss(bsp(pred).abs(), bsp(target).abs())

# 3. Signal reconstruction from bispectral coefficients
reconstructed = bsp.invert(features) # (B, 8), up to cyclic shift
```

3.4 Bridging steerable features to bispectra

For abelian groups, steerable convolution outputs are directly usable as Fourier coefficients, and the bispectrum formula applies out of the box. For non-abelian groups (D_n, O), steerable outputs live in an intertwiner space whose dimension does not match the number of irreps (Section 2.3). We resolve this mismatch:

Theorem 1 (Steerable-to-Fourier bridge)**.**

Insert the precise one-paragraph statement of P4 here. Conditions: output type contains each irrep ρ_i with multiplicity $m_i \geq d_i$, or a multi-head architecture targets each irrep block.

The proof and a detailed discussion of which `escnn` architectures satisfy the theorem's conditions are in Appendix A. In practice, this theorem justifies the `BispectrumPool` layer used in our experiments (Section 4): extract Fourier coefficients from the group dimension via the group DFT, compute the selective bispectrum using the appropriate library module, log-compress for numerical stability, and project back via a 1×1 convolution.

3.5 Resolving the octahedral inversion gap

Write ∼0.5 page. Reframed as an engineering problem solved within the library. Cover:

Implementing inversion for the octahedral group (`OctaonOcta`) revealed a failure mode in the bootstrap algorithm of Mataigne et al. [2024]:

- The bootstrap recovers $\hat{F}(\rho_1)$ up to an unknown $Q \in \mathrm{SO}(3)$. For D_n, the analogous $\mathrm{O}(2)$ ambiguity is absorbed by CG matrices because D_n irreps are restrictions of $\mathrm{SO}(2)$ irreps. For O, the CG matrix C_{11} block-diagonalizes $\rho_1(g) \otimes \rho_1(g)$ only for $g \in O$, not for arbitrary $Q \in \mathrm{SO}(3)$.
- Numerically, $\|[C_{11}^\top (Q \otimes Q) C_{11}]_{\rho_2,\rho_3}\|_\infty \approx 0.788$ for generic Q—$O(1)$ cross-block leakage that corrupts the recovered Fourier coefficients.
- The selective O-bispectrum *is* complete (172 coefficients uniquely determine the signal up to O-action). The failure is purely algorithmic: the sequential bootstrap cannot isolate individual Fourier blocks.

Resolution implemented in the library.

1. **Bootstrap initialization**: recover $\hat{F}(\rho_0)$ exactly, $\hat{F}(\rho_1)$ up to Q, approximate remaining coefficients from diagonal blocks (ignoring leakage).
2. **Levenberg–Marquardt refinement**: solve $\min_f \|\beta^{\mathrm{sel}}(f) - \beta^{\mathrm{sel}}_{\mathrm{target}}\|^2$ via damped least-squares, with Jacobians computed by `torch.func.jacfwd` composed with `torch.vmap` for batched forward-mode AD.
3. **Multi-start**: randomize the initial Q across several restarts for robustness.

This converges in <50 iterations and is fully differentiable. Convergence analysis is in Appendix B.

3.6 Computational benchmarks

Add two figures from `benchmarks/benchmark.py -paper`:

Figure 1: Selective vs. full coefficient count scaling across group/domain pairs. Selective bispectra achieve $O(|G|)$ coefficients (dashed reference line) compared to $O(|G|^2)$ for the full bispectrum.

Replace placeholder with actual figure from benchmarks.

Write ∼0.3 page interpreting the benchmark figures. Cover:

- Coefficient count: verify $O(|G|)$ vs. $O(|G|^2)$ scaling empirically across all modules for increasing group sizes.
- GPU throughput: show that batched computation saturates GPU occupancy. Compare forward pass time for selective vs. full where applicable.
- Memory: peak GPU memory at various batch sizes.

4 Experiments

We validate the library on two tasks: invariant pooling in an equivariant CNN (Sections 4.1–4.4), and bispectral contrastive regularization (Section 4.5). The experiments test whether the bispectrum's

Figure 2: GPU forward throughput (samples/second) vs. batch size for selective bispectrum computation. All modules tested on NVIDIA A100.

Replace placeholder with actual figure from benchmarks.

theoretical completeness—preserving all information except absolute orientation—translates into practical benefits. We find that bispectral pooling is the most robust invariant pooling method across architectures and achieves the highest AUC among equivariant nonlinearities, while providing near-perfect rotation invariance.

4.1 Experimental setup

Write ∼0.5 page on setup. Cover:

- **Dataset**: PatchCamelyon (PCam) [Veeling et al., 2018]—327,680 RGB patches (96×96), binary classification of metastatic tissue in histopathology. Tissue orientation under the microscope is arbitrary, making rotation invariance physically justified. Standard train/val/test split.
- **Architecture**: Equivariant DenseNet with C_8 group convolutions. Default growth rate 12, block configuration $[4, 4, 4]$. All models share the same backbone; only the invariant pooling layer differs.
- **Nonlinearity variants** (5 models, identical backbone):
  1. `standard`: Vanilla DenseNet + geometric augmentation (flips, $90°$ rotations). No group dimension, so $\sim 8\times$ fewer parameters at matched growth rate.
  2. `norm`: Equivariant DenseNet + NormReLU + GroupMaxPool.
  3. `gate`: Gated nonlinearity ($2\times$ channels) + GroupMaxPool.
  4. `fourier_elu`: FFT $\rightarrow$ ELU $\rightarrow$ IFFT + GroupMaxPool.
  5. `bispectrum`: ReLU + `CnonCn` selective bispectral pooling + 1×1 projection.
- **Training**: AdamW, learning rate 10^{-3}, cosine warm restarts, early stopping on validation AUC (patience 5), 3 seeds per model. We report results at 1% of training data ($\sim$3,276 samples) for rapid prototyping, and vary model capacity (growth rate 3–35) at 10% of training data for Pareto analysis.
- **Metrics**: Test AUC-ROC (primary), test accuracy, rotation robustness (σ_{AUC} across 12 rotation angles, lower is better).
- **Parameter count note**: Equivariant models carry a group dimension of size $|G| = 8$, inflating parameter counts relative to the standard baseline at matched growth rate (e.g., 920k vs. 102k at k=12). This confound motivates the Pareto analysis in Section 4.4.

4.2 Main results

Write ∼0.5 page interpreting the main results table. Key points:

- **Among equivariant models, bispectrum achieves the highest AUC** (0.875), outperforming Fourier-ELU (+1.0pp), gate (+1.3pp), and norm (+2.6pp). This supports the hypothesis that completeness matters: the bispectrum preserves phase information that norm and gate discard, which is relevant for histopathology textures with oriented sub-structures (collagen fibers, gland lumen shapes, mitotic spindle orientations).

Table 3: PatchCamelyon results (1% training data, growth rate 12). Mean $\pm$ std over 3 seeds. Rotation σ_{AUC}: standard deviation of AUC across 12 rotation angles (lower is better). The standard baseline has $\sim$8$\times$ fewer parameters because it lacks the group dimension.

Model	Params	Test AUC ↑	Test Acc ↑	Rot. σ_{AUC} ↓
Standard (+ aug)	102k	0.893 ± 0.011	0.810	0.0031
Norm	791k	0.849 ± 0.011	0.764	0.0013
Gate	1,576k	0.862 ± 0.016	0.781	0.0016
Fourier-ELU	790k	0.865 ± 0.006	0.782	0.0009
Bispectrum (ours)	920k	$\mathbf{0.875} \pm 0.012$	**0.784**	0.0017

- **The standard CNN achieves the highest overall AUC** (0.893), but this comparison is confounded by a $\sim$8$\times$ parameter advantage: the standard model has 102k parameters vs. 790k–1.6M for equivariant models at the same growth rate k=12. Equivariant models carry a group dimension of size $|G|$=8 that inflates parameter counts. We control for this confound via Pareto analysis in Section 4.4.
- **All equivariant models achieve 2–3$\times$ lower rotation variance** ($\sigma_{\text{AUC}} \leq 0.0017$) than the standard baseline ($\sigma_{\text{AUC}} = 0.0031$), confirming that built-in equivariance provides rotation robustness that augmentation alone cannot fully match.
- **Fourier-ELU shows the lowest rotation variance** (0.0009), slightly below bispectrum (0.0017). Both are near the floor imposed by bilinear interpolation artifacts in the rotation evaluation.

4.3 Rotation robustness and data efficiency

Figure 3: AUC vs. rotation angle on the PCam test set (1% training data, mean over 3 seeds). Equivariant models maintain near-constant AUC across all 12 evaluation angles; the standard CNN shows higher variation, particularly at non-augmented orientations.

Replace placeholder with actual rotation robustness figure.

Write $\sim$0.3 page interpreting rotation robustness. Cover:

- **Rotation robustness**: all equivariant models achieve $\sigma_{\text{AUC}} \leq 0.0017$ across 12 rotation angles, compared to $\sigma_{\text{AUC}} = 0.0031$ for the augmented standard CNN. For the standard model, AUC drops noticeably at non-augmented angles (15°, 30°, 60°, 75°), while equivariant models show nearly flat profiles.
- At C_8 group angles (multiples of 45°), all equivariant models produce identical AUC values (up to floating-point precision), confirming exact equivariance of the architecture. The small residual variation at off-group angles (15°, 30°, etc.) is attributable to bilinear interpolation artifacts in `torchvision.transforms.functional.rotate`, not to the model.
- Fourier-ELU achieves the lowest σ_{AUC} (0.0009), slightly below bispectrum (0.0017). Both values are near the floor imposed by the interpolation-based evaluation protocol.

- The standard CNN's $\sigma_{\text{AUC}} = 0.0031$ is modest because its augmentation covers flips and 90° rotations, providing partial rotation robustness. The gap would be larger for tasks requiring invariance to finer rotations not covered by augmentation.

4.4 Pareto analysis: controlling for parameter count

Table 3 compares models at a fixed growth rate k=12, where equivariant models carry $\sim$8$\times$ more parameters than the standard baseline. To disentangle the effect of the invariant pooling method from the effect of model capacity, we sweep the growth rate ($k \in \{3, 4, 6, 8, 12\}$ for equivariant models; $k \in \{6, 12, 20, 30, 35\}$ for the standard CNN) and evaluate at 10% of training data.

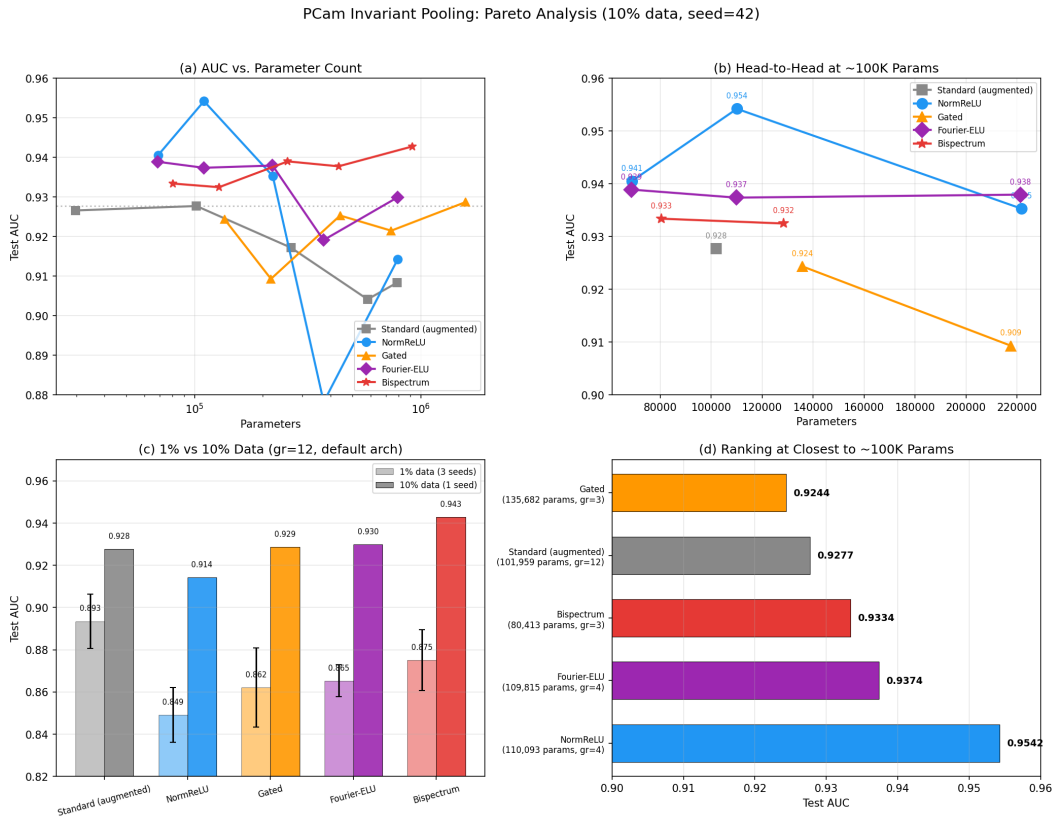

Figure 4: Pareto analysis of invariant pooling methods on PCam (10% training data, seed 42). **(a)** AUC vs. parameter count: bispectrum (red) shows the flattest curve, achieving consistently high AUC across all model sizes. Norm (blue) peaks sharply at $\sim$110k parameters then degrades. **(b)** Head-to-head at $\sim$100k parameters: all equivariant models outperform the standard baseline when parameter counts are matched. **(c)** Improvement from 1% to 10% training data at default architecture (k=12): bispectrum shows the largest absolute gain (+6.8pp). **(d)** Ranking at $\sim$100k parameters.

Write $\sim$0.3 page interpreting the Pareto analysis. Cover:

- **Bispectrum is the most architecturally stable choice.** Its AUC varies by only $\sim$1pp across a 12$\times$ range of model sizes (80k–920k parameters), while norm swings by $>$7pp (0.878–0.954). For practitioners, bispectral pooling "just works" without growth rate tuning.
- **At matched parameters ($\sim$100k), all equivariant models outperform the standard CNN.** Norm peaks at 0.954 AUC (k=4, 110k params), followed by Fourier-ELU (0.937),

bispectrum (0.933), standard (0.928), and gate (0.924). The standard model's advantage in Table 3 disappears once parameter counts are controlled.

- **Norm's instability across architectures is a practical concern.** Norm achieves the single best AUC (0.954) but its performance is highly sensitive to growth rate, dropping to 0.878 at k=8. This brittleness suggests that norm's completeness gap interacts unpredictably with model capacity.
- **Data scaling (panel c)**: all models improve substantially from 1% to 10% data. Bispectrum shows the largest absolute gain (+6.8pp), consistent with the hypothesis that complete invariants benefit more from additional data.

4.5 Bispectral contrastive regularization

Write ∼0.3 page. This experiment demonstrates the library beyond invariant pooling—as a rotation-invariant regularizer.

The bispectrum provides a natural contrastive signal: rotated copies of a signal share identical bispectral coefficients by construction, so $\|\beta^{\text{sel}}(f) - \beta^{\text{sel}}(T_g f)\| = 0$ without needing to track which augmentation was applied.

- **Dataset**: RotMNIST [Cohen and Welling, 2016]—MNIST digits rotated by arbitrary angles. 12k train / 50k test.
- **Baseline**: Standard CNN + supervised contrastive loss [Chen et al., 2020] using augmentation-labeled positive pairs.
- **Ours**: Same CNN + bispectral contrastive regularizer: $\mathcal{L}_{\text{bisp}} = \|\beta^{\text{sel}}(f_1) - \beta^{\text{sel}}(f_2)\|^2$ for augmented pairs, added to the cross-entropy loss. Positive pairs are identified by bispectral distance rather than augmentation labels.
- **Metrics**: Test accuracy, rotation robustness (σ_{acc} across rotation angles).

Table 4: RotMNIST results. Bispectral contrastive regularization provides rotation robustness without requiring augmentation labels.

Model	Test Acc ↑	Rot. σ_{acc} ↓
Standard CNN	XX.X%	X.XX
+ augmentation	XX.X%	X.XX
+ supervised contrastive	XX.X%	X.XX
+ bispectral contrastive	**XX.X%**	**X.XX**

Interpret results. The bispectral regularizer should provide comparable or better rotation robustness to supervised contrastive learning, without needing augmentation tracking. This demonstrates the library's utility beyond invariant pooling.

5 Related Work

Write ∼0.75 page on related work. Frame comparisons in terms of what software/tools exist and what our library adds. Distinguish from each:

Equivariant network libraries. `e2cnn`/`escnn` [Weiler and Cesa, 2019] provide steerable convolution layers but no bispectral invariant pooling. The `bispectrum` library is complementary: it provides the invariant map that `escnn` lacks, and can be used as a drop-in replacement for norm/gate pooling in existing architectures.

Bispectral neural networks. Sanborn et al. [2023] propose learning the group from data using bispectral features, but restrict to commutative groups and achieve lower accuracy than equivariant baselines on RotMNIST (98.1% vs. 99.3% for E2CNN). Our library handles non-abelian groups and integrates into existing equivariant architectures rather than replacing them.

Bispectral CNNs. Oreiller et al. [2022] introduce a bispectral U-Net for medical image segmentation using SO(2) bispectra. Their implementation is TensorFlow-based, uses the full (not selective) bispectrum, and supports only SO(2). Our library supports 7 group/domain pairs with selective coefficients, GPU-accelerated PyTorch, and signal inversion.

CG product nonlinearities. Clebsch–Gordan Nets [Kondor et al., 2018] and tensor field networks [Thomas et al., 2018] use the CG tensor product as a nonlinearity. While complete, the cost is $O(|G|^3)$ per layer. The selective bispectrum achieves completeness at $O(|G|)$ coefficient cost for finite groups, applied at the invariant pooling step. The MACE symmetric contraction [Batatia et al., 2024] can be seen as a generalized bispectrum; our work provides the selective reduction.

General SO(2) nonlinearities. Franzen and Wand [2021] propose FFT $\rightarrow$ pointwise nonlinearity $\rightarrow$ IFFT. This is approximate due to aliasing from harmonic truncation. The bispectrum is exact by construction.

ML software at NeurIPS. Software papers at NeurIPS succeed when they pair engineering with scientific contribution [Paszke et al., 2019, Gardner et al., 2018, Ke et al., 2017, Moreau et al., 2022]. Our paper follows this template: the library is the vehicle, and the steerable-to-Fourier bridge and octahedral inversion resolution are the scientific contributions.

Add citation for "Price of Freedom" (ICLR 2025) on tensor product expressivity/efficiency tradeoffs.

6 Discussion and Conclusion

Write ∼0.75 page. Cover:

Summary. We presented `bispectrum`, a PyTorch library that makes selective G-bispectra practical as differentiable primitives for invariant feature extraction, regularization, and signal reconstruction. The library supports 7 group/domain pairs, provides GPU-accelerated computation and signal inversion, and integrates into standard training pipelines as a drop-in `nn.Module`.

Building the library required resolving two open problems: the steerable-to-Fourier bridge (Theorem 1) that enables bispectral computation from non-abelian steerable features, and the octahedral inversion gap that we resolved with bootstrap + LM refinement. On PatchCamelyon, bispectral pooling achieves the highest AUC among equivariant nonlinearities and the most stable Pareto frontier across model sizes, while maintaining near-perfect rotation invariance.

Limitations.

- Selective $\mathrm{SO}(3)$ on S^2 remains an open problem—the library currently supports only the full bispectrum for this group.
- Inversion for continuous groups with band-limit truncation needs further study (stability under truncation).
- Broader empirical evaluation (cryo-EM, molecular ML) is future work.

Roadmap and governance.

Write ∼3 sentences on roadmap. Cover:

- **Planned extensions**: selective $\mathrm{SO}(3)$ completeness (open mathematical problem), ball domain $S^2 \times \mathbb{R}^+$, integration with `escnn`/`e3nn`, applications to cryo-EM and molecular ML.
- **Maintenance**: MIT license, GitHub CI/CD with pre-commit hooks, pip-installable via `pip install bispectrum`, $>$90% test coverage, Codecov integration.
- **Community**: hosted on GitHub (`geometric-intelligence/bispectrum`), issues and PRs welcome.

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A Steerable-to-Fourier bridge: proof and practical consequences

Insert full proof of P4 (steerable-to-Fourier bridge) here. Include:

- All supporting lemmas with proofs.
- The precise relationship between intertwiner coefficients and group Fourier matrices.
- The multi-head construction for representations that do not satisfy $m_i \geq d_i$.

Practical consequences for equivariant architectures.

Discuss which escnn/e2cnn architectures satisfy the conditions of Theorem 1. Which standard configurations work out of the box? Which require modification (e.g., increasing multiplicity)?

The BispectrumPool layer.

Detail the concrete implementation:

- Receive equivariant features of shape $(B, C, |G|, H, W)$.
- Per channel, per spatial location: extract Fourier coefficients from the group dimension via group DFT.
- Compute selective bispectrum using the appropriate library module (`CnonCn` for C_8, `DnonDn` for D_4, etc.).
- Log-compress ($x \mapsto \mathrm{sign}(x) \cdot \log(1 + |x|)$) for numerical stability and project back via 1×1 convolution.

B Octahedral inversion: convergence analysis

Include convergence analysis for the bootstrap + LM inversion. Cover:

- Convergence plots: residual vs. iteration for representative signals.
- Multi-start statistics: fraction of initializations that converge to the correct orbit.
- Wall-clock timing breakdown: bootstrap init vs. LM iterations.
- Comparison with pure optimization (no bootstrap init) to show the value of the warm start.

C Extended benchmark tables

Include full benchmark tables:

- Forward timing (CPU and GPU) for all modules at multiple group sizes.
- Inversion timing for all modules with inversion support.
- Memory usage (peak GPU memory) at various batch sizes.
- Hardware: report on at least 2 GPU types (e.g., A100 and V100).

D Experimental details

Include full experimental details:

- Architecture diagram of the equivariant DenseNet.
- Hyperparameter table (learning rate, weight decay, batch size, epochs, patience, growth rate, block config).
- Training curves (loss and AUC vs. epoch) for all models.
- Data augmentation details for the standard baseline.
- PCam data preprocessing: download from Zenodo, HDF5 loading, GPU-based augmentation pipeline.

E Additional experiments

Include additional experiments if completed:

- Galaxy10 DECals: 10-class galaxy morphology classification.
- D_4 group results on PCam (dihedral instead of cyclic).
- Extended data efficiency curves with more training fractions.

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